TRAFFIC CONGESTION PREDICTION

Project README

1. Project Overview

Traffic Congestion Prediction is a machine learning project that estimates the number of vehicles based on date, time, day-of-week, weekend, and junction inputs. The application is built using Python and Streamlit and provides a simple interface for users to enter details and view a prediction.

2. Objectives

• Estimate vehicle count from selected date and time features.

• Build a simple, user-friendly web application.

• Demonstrate the use of machine learning for a traffic-related problem.

• Provide a starting point for future traffic analysis and monitoring features.

3. Technologies Used

Python • Pandas • Scikit-learn • Joblib • Streamlit • Excel dataset

4. Input Features

The app accepts Year, Month, Day, Hour, Day of Week, Weekend (0 = No, 1 = Yes), and Junction.

5. Project Workflow

• Load the traffic dataset.

• Prepare date and time features for model training.

• Train a machine learning regression model to estimate vehicle count.

• Save the trained model in a .pkl file.

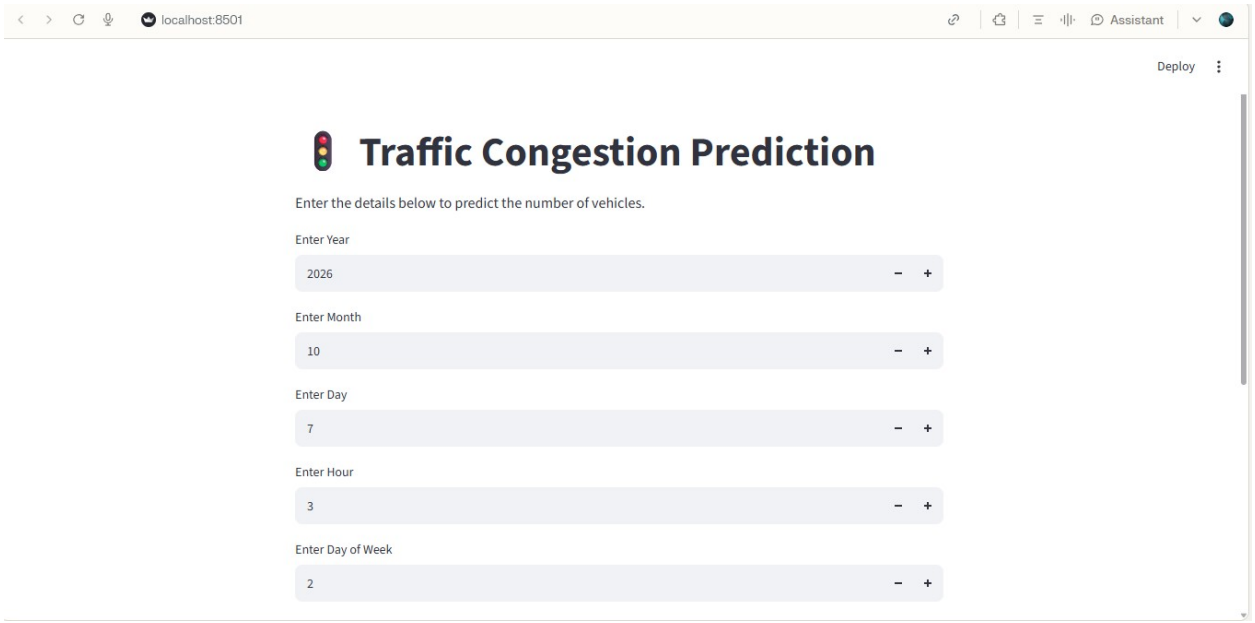
• Load the model in the Streamlit app.

• Accept input values and display the predicted number of vehicles.

6. Application Screenshots

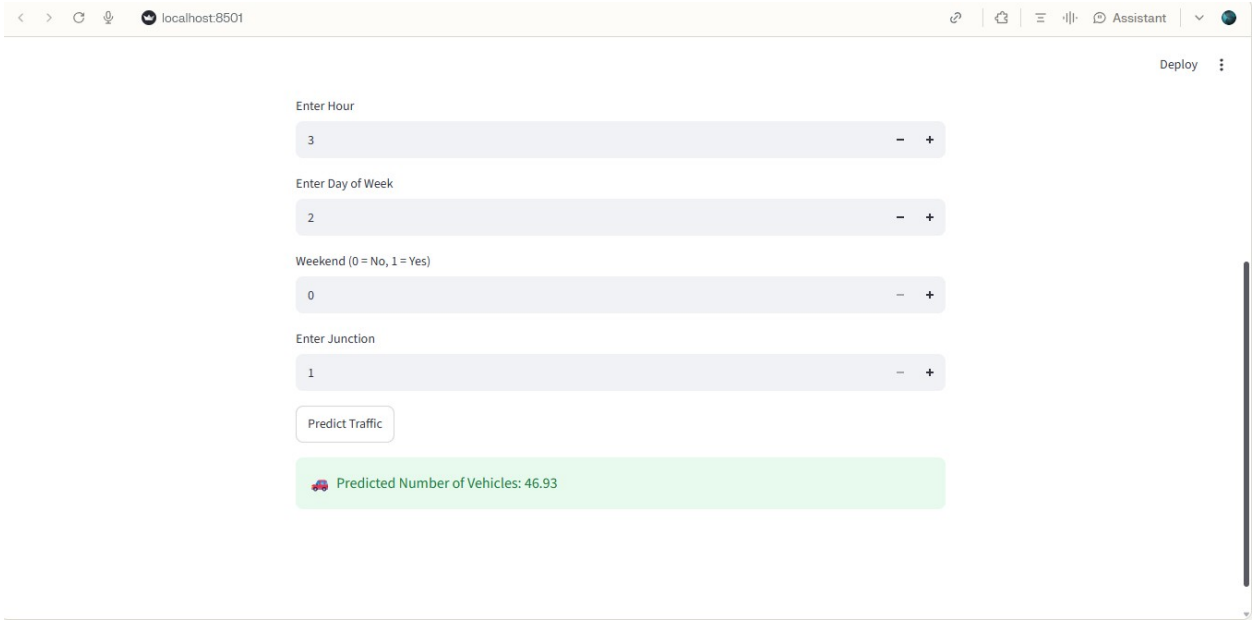
Input Form

Enter the date, time, day-of-week, weekend, and junction details.



Prediction Result

Example result: Predicted Number of Vehicles: 46.93. This is a model estimate, not a live sensor count.



7. How to Run the Project

Step 1 — Install the required packages

```
python -m pip install pandas scikit-learn joblib streamlit openpyxl
```

Step 2 — Keep the application and model files together

Example folder structure: Traffic-Congestion-Prediction/ with Trafficc.py, traffic_model.pkl, README.md, and screenshots/input_form.jpg plus screenshots/prediction_output.jpg. Use your actual Python script filename if different.

Step 3 — Run the Streamlit application

```
python -m streamlit run Trafficc.py
```

Replace Trafficc.py with your actual script filename if needed. Open the local URL shown in the terminal.

8. Output

The application displays the estimated number of vehicles for the entered feature values. The supplied screenshot shows an example result of 46.93 vehicles.

9. Future Enhancements

• Show low, medium, and high traffic levels using defined thresholds.

• Add charts for vehicle-count trends by hour, day, or junction.

• Evaluate and document model performance with regression metrics.

• Integrate live traffic sensor or traffic API data if available.

10. Limitations

The current interface shows a machine learning estimate based on the model and input features. The screenshots do not indicate a live traffic feed, so the displayed result should not be treated as a real-time vehicle count. Prediction quality depends on the training dataset and model performance.

11. Author

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